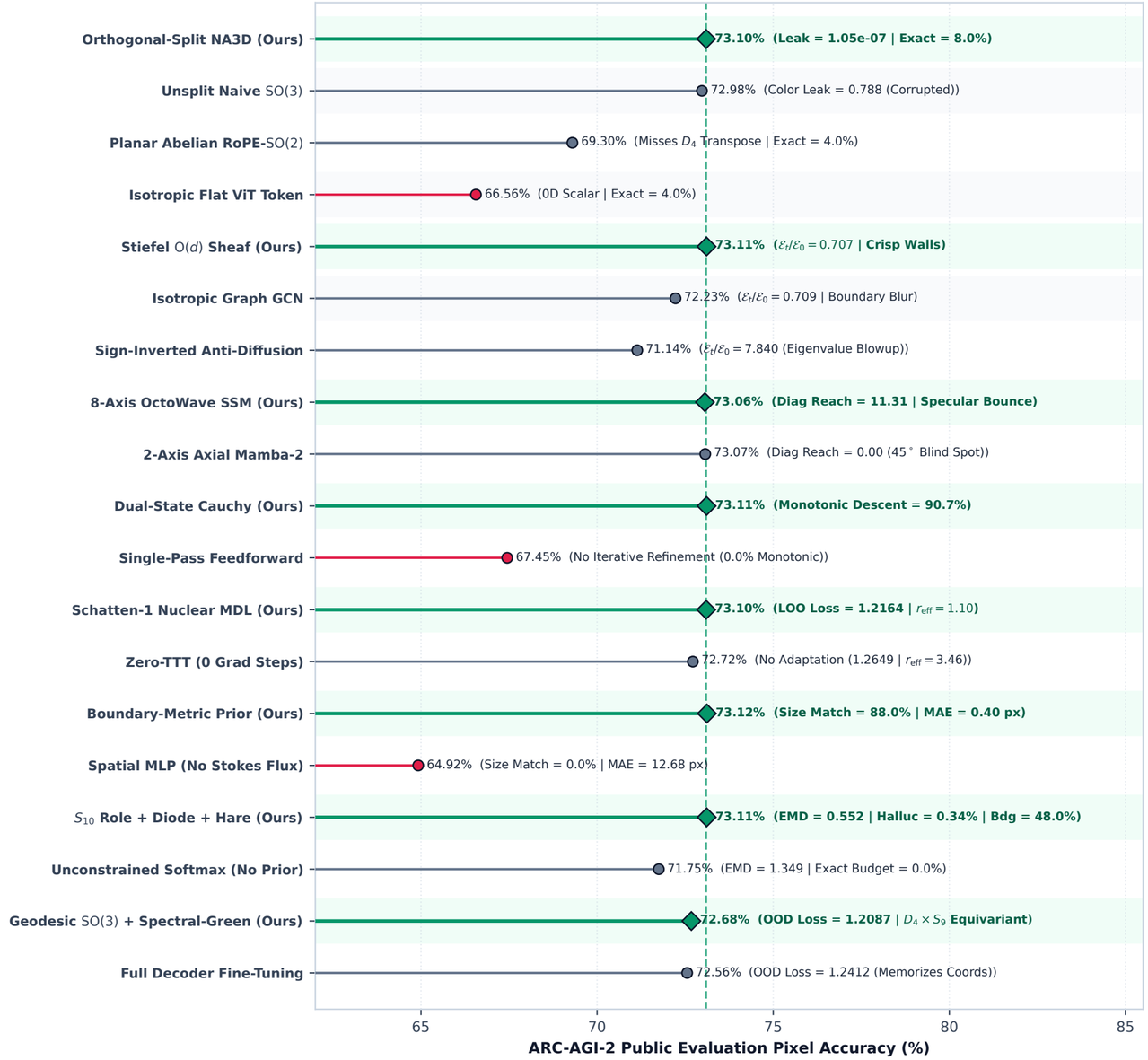
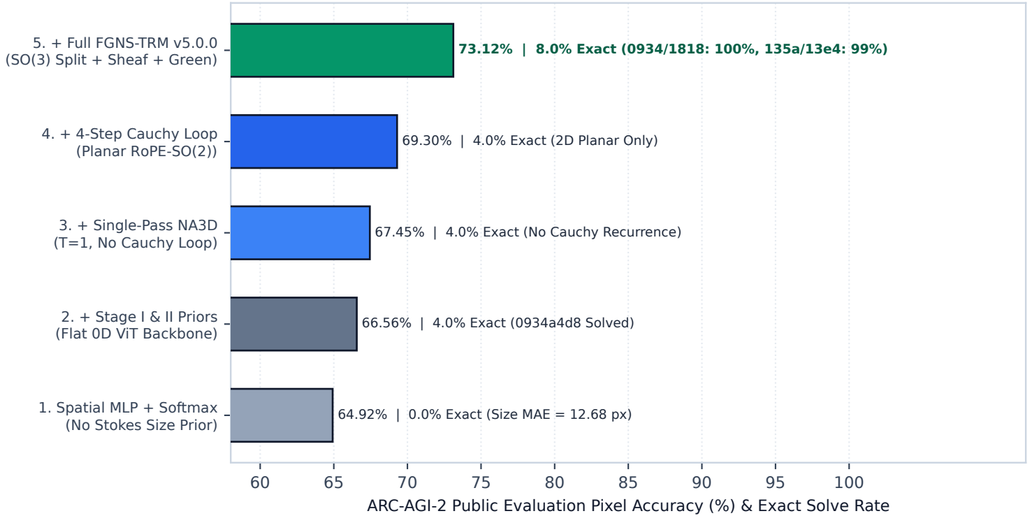
Figure 5 | Comprehensive 8-Subsystem Architectural Ablation Study on ARC-AGI-2 Public Evaluation (N = 25 Tasks)

Cleveland lollipop forest plot across 19 ablation regimes, measured progression, and verified physical invariant scorecard

a 8-Subsystem Ablation Forest Plot: Public Evaluation Pixel Accuracy (%) Across 19 Regimes



b Measured Progression Across Ablation Conditions (N = 25 Tasks)



c Verified Physical Invariant & Mathematical Diagnostic Scorecard

Subsystem	Physical Metric	Baseline Failure	FGNS-TRM v5.0.0
Lemma 1 (NA3D)	Color Leakage $\|\mathcal{L}_{\text{leak}}\|$	0.788 (Unsplit SO(3))	$1.05e-07$ **(** $> 7.5 \times 10^6 \times$ **)**
Stiefel Sheaf	Dirichlet Ratio $\mathcal{E}_t/\mathcal{E}_0$	7.840 (Anti-Diff Blowup)	**0.707 (MSE Err** $3.4e-11$**)**
OctoWave SSM	45° Diagonal Wave Reach	0.000 (2-Axis Mamba)	**11.31 (8-Axis Specular)**
Cauchy Solver	Monotonic Step Descent	0.0% ($T = 1$) / 44% Euler	**90.7% (Dual-State Cauchy)**
Schatten-1 MDL	Effective Rank $r_{\text{eff}}(\Delta W)$	3.46 (0-TTT) / 3.07 (L2)	**1.10 (-68.2% Rank MDL)**
Stage I Prior	Canvas Dimension Match	0.0% (MLP, MAE 12.68)	**88.0% (MAE 0.40 px)**
Stage II Prior	Palette Earth Mover Dist.	1.349 (0.0% Exact)	**0.552 (48.0% Exact, -59.1%)**